

IPL Data Analysis using Power BI

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Introduction

The Indian Premier League (IPL) is one of the most popular cricket leagues globally. The purpose of this project is to analyse IPL match data using Power BI and create an interactive dashboard for extracting meaningful insights.

The dashboard helps users understand:

- Team performances
- Player performances
- Toss impact on match results
- Venue and city analysis
- Batting and bowling statistics
- Match trends and season insights

The dashboard was built using:

- Power BI Desktop
- IPL Matches Dataset
- IPL Deliveries Dataset
- DAX Measures
- Data Cleaning using Power Query

Business objective

The main objectives of this project are:

1. To convert raw IPL data into meaningful business insights.
2. To identify top-performing teams and players.
3. To analyze the effect of toss decisions on match outcomes.
4. To identify lucky venues and city-wise match distribution.
5. To improve sports analytics understanding through data visualization.
6. To enable quick and interactive exploration of IPL trends.

Business Problems

The IPL management, teams, analysts, and cricket enthusiasts often face challenges in understanding large volumes of match data and extracting meaningful insights manually.

The major business problems identified in this project are:

1. Difficulty in analysing team performance across multiple IPL seasons.
2. Understanding factors behind successful IPL teams.
3. Lack of centralized insights regarding batting and bowling performances.
4. Inability to identify the impact of toss decisions on match outcomes.
5. Difficulty in identifying venue advantages and city-wise match trends.
6. Large raw datasets make manual analysis time-consuming.
7. Teams and analysts require quick decision-making insights through visual analytics.
8. Understanding player consistency and match-winning contributions is challenging without proper dashboards.

Data set information

Two datasets (2008-2017) were used in this project.

A. Matches Table

Contains:

- match_ID
- season
- teams
- toss_winner
- toss_decision
- match_winner
- venue
- city
- result

B. Deliveries Table

Contains:

- match_ID
- over
- ball
- batter
- bowler
- batsman_runs
- total_runs
- player_dismissed

DATA CLEANING

Data cleaning was performed in Power Query Editor.

1. Imported data
The IPL datasets were loaded into Power BI → using Get data → text/csv
2. Check data types
Date columns were set to Date format, numeric columns (runs, wickets) to Whole Number, and team/player names to Text.
3. Removed null values
Blank/null values were identified and removed from important columns to prevent them from affecting DAX formulas and calculations.
4. Corrected player_dismissed column
*Empty cells were removed with null values.
This corrected the wicket calculations.*
5. Removed duplicate rows
From eg. match_id, so it doesn't increase counts and create false insights.
6. Standardised team names
Rename the Delhi Daredevils with Delhi capitals
7. Remove unnecessary columns
Umpire3 column was removed to simplify the analysis.
8. Established relationship

`matches[id] → deliveries[match_id]`

Relationship type:

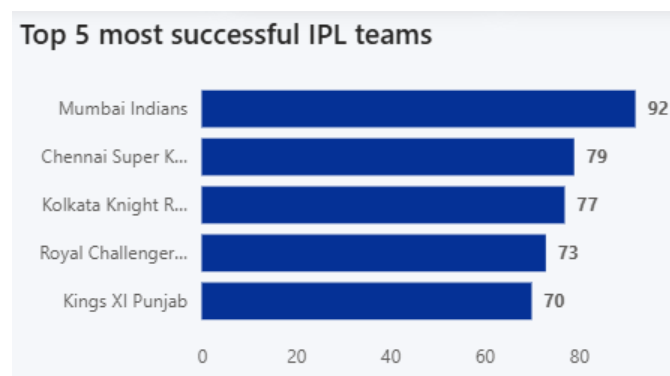
- One-to-many (1*)
 - Single cross filter
9. DAX measures were used to make calculations

DASHBOARD

KPI cards were created:

1. Total Matches
The IPL datasets contain over 633 matches, showing the league's large-scale and consistent growth over multiple seasons.
2. Total Runs
More than 190k runs were scored in the IPL, highlighting strong batting dominance.
3. Total Wickets
Bowlers still played a major role despite batting-friendly conditions, with over 7k wickets taken over seasons.
4. Total Sixes & Fours
Large numbers of fours and sixes indicate aggressive batting trends.
5. Total Team
A total of 14 teams participated across seasons.

MOST SUCCESSFUL TEAMS



Insights:

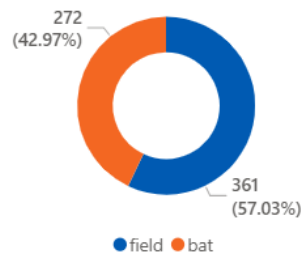
- Mumbai Indians has the best successful win record in the IPL.
- Chennai Super Kings and Kolkata Knight Riders are also highly consistent.

Business Insights

Strong team management, experienced leadership and better decision-making increase win consistency.

TOSS DECISION DISTRIBUTION

Toss decisions distribution



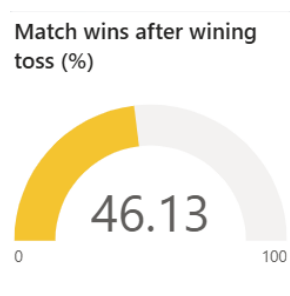
Insights:

- Most teams prefer fielding after winning the toss.
- Chasing is considered easier in T20 cricket, especially because teams understand targets better during the second innings.

Strategy Insight

Teams analyse pitch conditions and dew factor before choosing.

MATCH WINS AFTER WINNING TOSS



Insights:

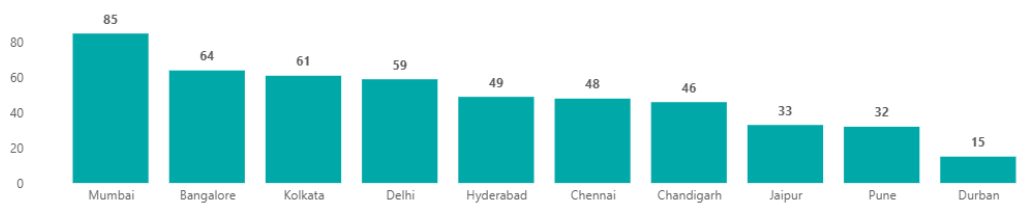
- Around 46% of teams that won toss also won the match.
- Toss helps slightly but does not guarantee victory.

Key conclusion

Team performance, player execution, and match situations matters more than toss advantage.

MATCHES HOSTED BY CITY

IPL Matches hosted by City



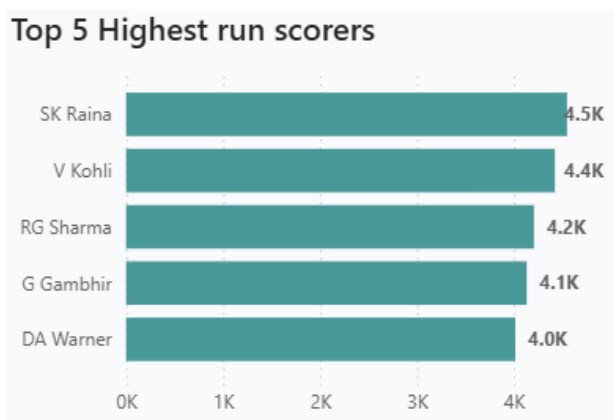
Insights:

- Mumbai hosted the highest number of IPL matches due to its strong cricket infrastructure, large stadiums, and huge fan base.
- Major cities continue to dominate hosting responsibilities because they generate higher ticket sales and viewership.

Business insights

IPL generates higher revenue in large cricket markets.

TOP 5 HIGHEST RUN SCORERS

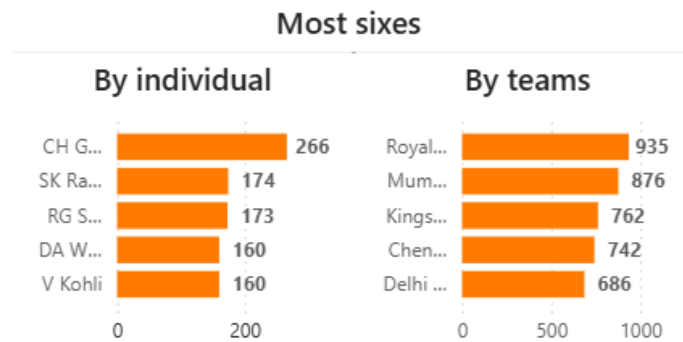


Insights:

- Suresh Raina and Virat Kohli are among the most consistent batsmen across multiple seasons.
- Consistency across seasons is more valuable than short-term performance.
- Their success comes from adaptability, consistency, and the ability to handle pressure situations.

Cricket insight

Top batsmen rotate strike effectively and maintain partnerships.



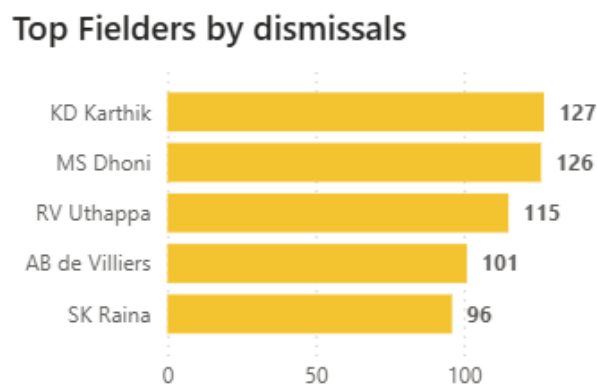
Insights:

- Chris Gayle dominates power-hitting statistics, showing the importance of explosive batting in T20 cricket.
- Teams like Royal Challengers Bangalore and Mumbai Indians rely heavily on aggressive batting lineups.

Strategy insight

Strategy focuses on maximising runs during powerplay and death overs.

TOP FIELDERS BY DISMISSALS



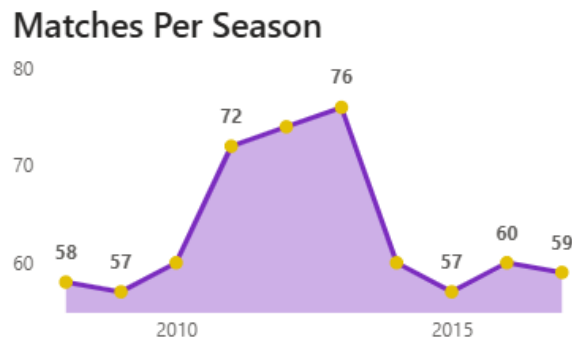
Insights:

- MS Dhoni, DK Karthik, and AB de Villiers contribute heavily through catches and run-outs.
- Good fielding creates pressure on opponents and often changes the match momentum.

Business Insight

Strong fielding reduces opposition scoring opportunities.

MATCHES PER SEASON TREND



Insights:

- The number of IPL matches increased significantly after league expansion.
- As IPL popularity grew, the league expanded its schedule and commercial reach.
- As seasons progressed, match margins reduced, player quality improved and team strategies become more data-driven.

RECOMMENDATIONS

For teams:

- Invest in death-over bowlers because wicket-taking ability strongly influences T20 match outcomes.
- Build stronger fielding units.
- Teams should optimize home-ground strategies because familiar pitch conditions improve win probability.
- Use data analytics for toss and venue strategies.
- Prioritise consistent players over short-term performers.

For Business

- Expand matches in high-revenue cities.
- Improve fan engagement in smaller venues.
- Schedule matches based on audience trends.

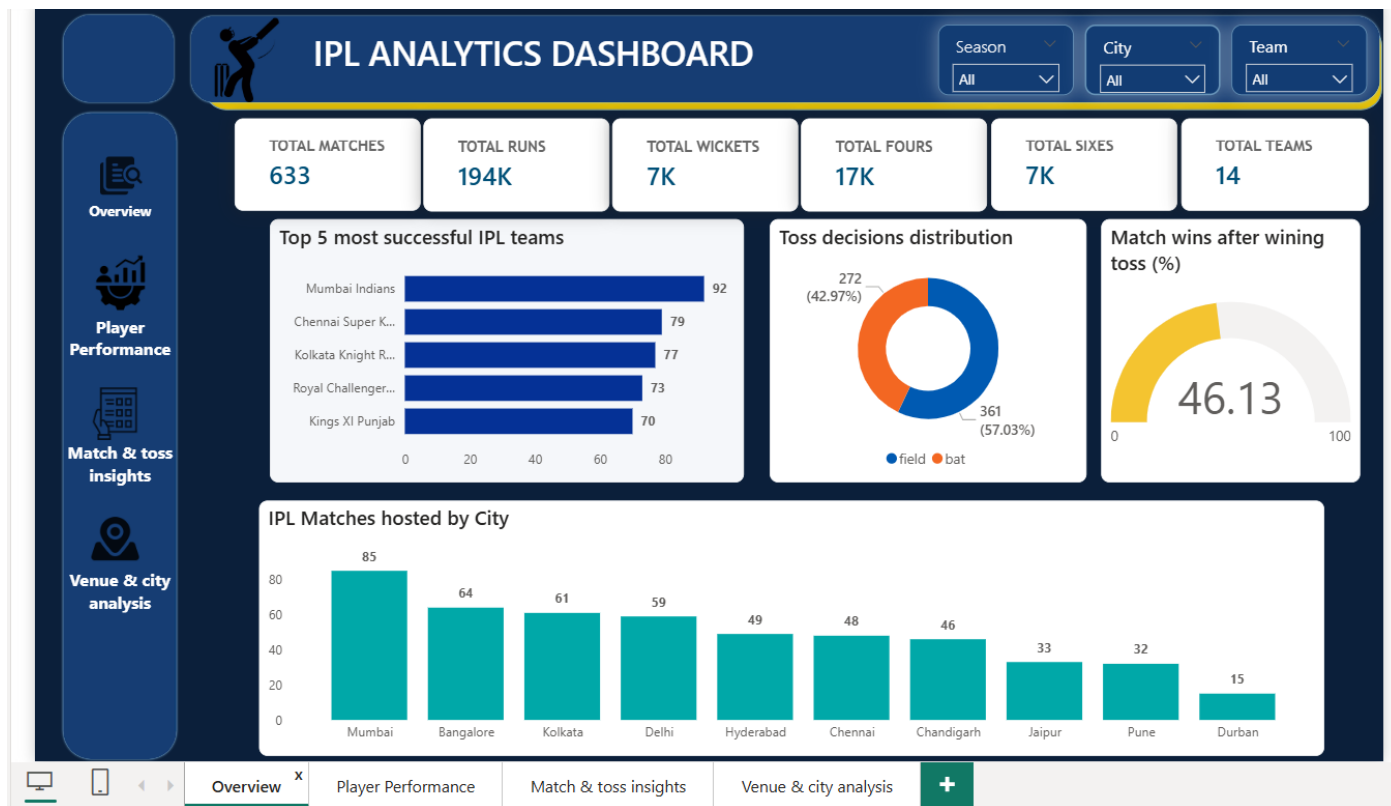
CONCLUSION

This IPL Analytics Dashboard successfully analyses team performance, player statistics, toss impact, and venue analysis using Power BI.

The analysis clearly shows that:

- Consistent teams perform better over long seasons.
- Toss advantage alone does not guarantee victory.

- Aggressive batting has become essential in T20 cricket.
- Strong bowling and fielding remain crucial for success.
- Venue familiarity and home advantage significantly influence match.



Overview

Player Performance

Match & toss insights

Venue & city analysis

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